

BEFORE THE
UNITED STATES COPYRIGHT OFFICE
LIBRARY OF CONGRESS

IN THE MATTER OF
SECTION 1201 STUDY:
REQUEST FOR ADDITIONAL COMMENTS
DOCKET 2015-0012

REPLY COMMENTS OF
THE REPAIR ASSOCIATION AND iFixIT

Submitted on behalf of
The Repair Association and iFixit by

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INTRODUCTION

About The Repair Association

The Repair Association, formerly known as the Digital Right to Repair Coalition, is a 501(c)6 trade association founded in July of 2013 to support individuals and businesses engaged in repair, reuse, and recycling of digital electronic parts and products. Our members range in size from individual hobbyists to multinational repair and services providers, and include consumer rights organizations, sustainability advocates, agricultural organizations, flexible labor networks, equipment trading networks, and logistical services networks.

The Repair Association has advised state governments on right to repair laws based on the 2012 Massachusetts Automotive Right to Repair law.¹ This bill provided a clear path for state legislative action, and also served as the foundation of major national voluntary agreements adopted by automotive² and commercial vehicle³ manufacturers. The success of this approach strongly suggests that commercial interests and consumer repair are complementary values that can both be served by well-considered policy.

About iFixit

iFixit.com is an international, open-source, online repair manual for everything. Our mission is to provide people with the knowledge they need to make their things work for as long as possible.

iFixit represents a global community of makers, tinkerers, fixers, and repair professionals. In 2015, the iFixit community taught repair to over 80 million people from almost every country in the world. The strongly collaborative group has published over 20,000 repair guides. This massive, free resource has helped people fix everything from cellphones to game consoles, toys to musical instruments. iFixit also stands firm in its

¹ “An Act protecting motor vehicle owners and small businesses in repairing motor vehicles”, Massachusetts Session Law of 2012, Chapter 368, *available at* <https://malegislature.gov/Laws/SessionLaws/Acts/2012/Chapter368>.

² Memorandum of Understanding, January 14, 2014, *available at* http://www.nastf.org/files/public/OtherReference/MOU_SIGNED_1_15_14.pdf.

³ Memorandum of Understanding, *available at* http://www.nastf.org/files/public/OtherReference/HD-MOU_2015.pdf.

support of the tinkerers and independent repair professionals in our community. We believe that owners should have the right to repair, modify, and tinker with the things that they own.

REPLY

The Repair Association and iFixit thank the Copyright Office for this opportunity to reply to the comments of the Association of Equipment Manufacturers (AEM) and the Equipment Dealers Association (EDA).⁴

The market in “commercially available” software

In their filing, the AEM and EDA argue against exemptions on a broad class of software, saying that:

“AEM and EDA caution against enactment of any permanent exemption where the copyrighted work, such as vehicle software or software updates, are commercially available products or services. If the works are already commercially available, the proposed permanent exemption would diminish or destroy the value of the vehicle software or software updates and unfairly disrupt the competitive landscape for both AEM and EDA members.” (p. 2)

This argument, and particularly these examples, presumes that software integrated into equipment is “commercially available to consumers.” This is not, and has never been, the case. Instead, this sort of software (often called embedded software or firmware) is included with the product. This is not surprising, since if it was not included, or was sold for an additional fee, the product would not work! As a result, vehicle software and software updates are not typically available separate from the hardware they power. This is particularly true in a world where, as we noted in our original filing, software is “eating the world” and therefore is included in essentially every product and category of product.

As a result, it is not surprising that there is no public market in firmware distinct from the market in the underlying products — you can’t browse to Amazon’s “firmware” section, or find a firmware aisle at Best Buy. Instead, the software produced by AEM’s members is typically offered free of charge to consumers. For example, one can visit the

⁴ Available at <https://www.regulations.gov/document?D=COLC-2015-0012-0105>.

Intel website and download firmware for any of its products at no charge.⁵ Similarly, Bosch is the leading manufacturer of Engine Control Units that are used within most motor vehicles worldwide. Bosch makes their firmware publicly available.⁶

Because of this availability, owners that need to return firmware to the original settings, or obtain updates, can do so easily, without any economic harm to manufacturers like Bosch, their retail distributors, or to future owners. The same type of firmware access is common for thousands of other components and presents no economic risk, nor any risk of piracy, to manufacturers. There is no economic value to firmware outside of the unique part and no value to preventing restoration or use.

This economic model for the production of firmware predates the DMCA, and will survive any broadened exceptions. First, as already noted, the economic driver for manufacturers is the sale of hardware, not software. This effect is strengthened by the nature of firmware, which is specific to particular pieces of hardware. Bosch ECU firmware, for example, will not work on ECUs made by other manufacturers, which both incentivizes Bosch to produce new firmware for new hardware, and essentially eliminates the supposed market discussed by AEM's comments. In addition, no-cost distribution of updated firmware is a form of safety and security defect support, and so this software will continue to be produced for those reasons – no additional protection, above and beyond that already offered by copyright law, is needed.

The real market in this area is not in selling software, but rather in going beyond software to limit competition in the market for repair, and to allow consumer-hostile behaviors that break devices owned by consumers without the consent or participation of their owners. For example, Apple recently shipped an iOS “update” that broke the iPhone for hundreds of thousands of users who had done nothing more sinister than replace broken physical screens.⁷ Similarly, a recent update to HP printers prevented them from operating with ink bought by a vendor other than HP.⁸ These “updates” suggest that AEM's interest is not in fostering a market for firmware (which would be

5 See, for example, Intel firmware available at <https://downloadcenter.intel.com/download/18363/Intel-SSD-Firmware-Update-Tool>.

6 See, for example, Bosch's firmware search engine at <http://www.bosch-automotive-catalog.com/en/vehicle-search>.

7 “Apple Apologizes And Updates iOS To Restore iPhones Disabled By Error 53”, *TechCrunch*, Feb. 18, 2016, available at <https://techcrunch.com/2016/02/18/apple-apologizes-and-updates-ios-to-restore-iphones-disabled-by-error-53/>.

impossible in any case) and rather in using software to impact policy and markets well outside the statutory and Constitutional purpose of copyright.

Public safety and the environment

Under the logic of the AEM/EDA response, copyright law should be used to regulate and enforce matters of public safety, environment and economic policy. In their response, as well as submissions to past triennial DMCA rulemakings, AEM/EDA propose that employees of manufacturers and dealerships help enforce federal policies by refusing to allow competition for repair. As AEM/EDA's comments read:

“... the restriction of this information to a finite number of highly qualified and readily identifiable parties *aids in the efficient and effective enforcement of safety and environmental regulations.*” (emphasis added) (p. 3)

When making this argument, AEM/EDA fail to cite to relevant statutory language, because there is none. But pursuing this route would not just be a violation of clear statutory and Congressional intent. It would also take the Office outside of its core competencies, inevitably creating inefficiencies both for the Office and for the newly regulated industries that would come under its purview.

The Copyright Office should reject this interpretation of its role, and leave the enforcement of safety and environmental regulations to the agencies best situated to do so.

“Economic disadvantage”

AEM/EDA's comments suggest that any exceptions to the DMCA will enable and encourage repair businesses to make unsafe modifications to firmware, and put “safe” providers at an economic disadvantage. (p. 1)

This position ignores the existence of a mass of federal, state, and local statutes and regulations that deal with safety and other issues. These rules have been created through the deliberate action of appropriate legislative, administrative, and judicial bodies. Refocusing enforcement of the Copyright Act on copyright concerns will not impact or lessen the application of those laws and regulations.

⁸ “HP: Disabling 3rd-party ink ensures ‘best printing experience’”, *Ars Technica*, Sep. 28, 2016, *available at* <http://arstechnica.com/information-technology/2016/09/hp-to-issue-optional-firmware-update-allowing-3rd-party-ink/>.

The use of electronic locks that prevent repair, or use of consumables such as printer ink⁹, are rarely, if ever, been about safety – instead, they have been about protecting the competitive position of manufacturers for repair services or consumable purchases. The “economic disadvantage” that AEM/EDA seeks to avoid is that of competition for goods and services provided by those who would seek to repair and improve the original products.

Where hardware manufacturers are actually providing firmware that meets the needs of consumers, new entrants will have no “unfair” advantage. Instead, limited by the same safety and environmental regulations as the original manufacturers, they will advance the “useful arts” through the basic rights of fair use, first sale, interoperability, and repair – rights that a broad interpretation of Section 1201 has taken away from them.

CONCLUSION

As software has come to be embedded in everything, Section 1201 has come to have a direct bearing on the right of every American, including both consumers and businesses, to repair and modify the devices they have purchased. The current review process has served mostly to show that existing exemptions, and the exemption process, are broken. For Americans to regain the right to repair their devices, they must be broadly reformed.

Respectfully submitted-

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⁹ *Id.*